

```

"truck": 2.5, "airplane": 36.0, "train":
3.2, "boat": 5.0, "traffic light": 0.6,

"fire hydrant": 0.3, "stop sign": 0.75,
"cat": 0.3, "dog": 0.6, "horse": 1.2,

"cow": 1.5, "elephant": 3.2, "bear": 1.7,
"zebra": 1.2, "giraffe": 2.0,

"bench": 1.2, "chair": 0.6, "couch": 2.0,
"dining table": 1.8,

"laptop": 0.4, "tv": 1.2
}
DEFAULT_OBJECT_WIDTH = 0.5 # Default
width (in meters) for unknown objects
FOCAL_LENGTH = 0.5 # Focal length in
meters (Raspberry Pi Camera Module 3 Wide
Spec)
To run the measure_distance3.py script
and capture data from different camera
sources:
$> python measure_distance3.py --input
rpicam # Capture data from Pi camera
$> python measure_distance3.py --input /
dev/video0 # Capture data from USB camera

```

The model yolov5n_seg_h8l_mz.hef supports around 80 object classes, but in this implementation, focus is only on 26 specific objects listed in the previous dictionary. These are the objects for which the standard physical widths have been provided, allowing for the estimation of their distance from the camera.

Testing

Fig. 3 illustrates the distance measurement technique, while Fig. 4 and Fig. 5 show pin details and mounting.

Power the system and run the script. Objects and their distances will be displayed in real time.

The Raspberry Pi 5, paired with the Hailo-8L AI HAT and powered by a 5V, 4A supply, is ideal for embedded or stationary applications that require real-time object detection and distance estimation.

This system can be used for various purposes, such as:

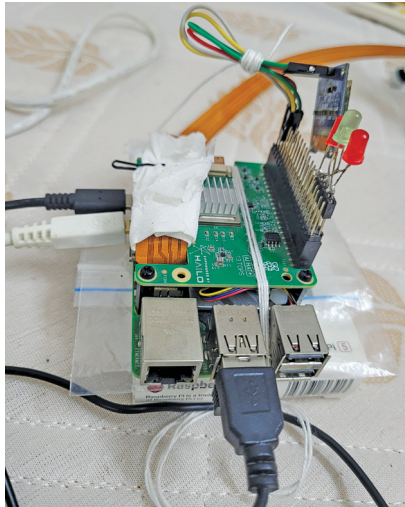


Fig. 5: Mounting of camera and Raspberry Pi

Car dashboard monitoring.

Mounted over the windshield, the system can detect vehicles and pedestrians ahead, estimate distances, and alert the driver in case of potential collisions or unsafe following distances. It can also integrate a speech synthesiser, such as espeak-ng, to issue voice advisories through a speaker.

Reverse gear monitoring with advisory. Installed at the rear of the car, it can detect nearby objects or people during parking and provide visual/audio alerts, or even trigger automatic braking mechanisms.

Smart door entry monitoring.

Positioned at an entrance, the system can detect approaching individuals, classify them (such as delivery persons, known visitors, or strangers), and estimate their distance to enable automated actions, including

doorbell activation, lighting, or door unlocking.

Occupancy and crowd analysis in buildings. It can be used to monitor the density and spacing of people in halls, corridors, or waiting areas. The system can trigger ventilation systems or crowd control alerts when individuals are too close to each other or the area becomes overcrowded.

Warehouse safety systems. It can detect moving forklifts or personnel and estimate their proximity to restricted zones. This can help trigger alarms or slow down automated machinery in the event of a near approach, thereby enhancing workplace safety.

Robot vision for indoor navigation. It is used in AGVs (automated guided vehicles) or indoor robots to identify obstacles and measure distances, enabling safer and more efficient navigation in indoor environments.

School zone or gate surveillance. The system monitors the presence and movement of children or guardians near school gates and generates alerts or automates gate operations based on detected behaviour and distance.

Traffic monitoring at toll booths or bridges. The system identifies vehicle types and measures distances to enable automated toll collection, detect overloading, or analyse traffic density.

School bus gate monitoring. It can help observe incoming traffic and the movement of children around school bus gates, ensuring safer entry and exit conditions. **EPF**

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